

ENERGY: OIL SERVICES

Key takeaways from call with Viridien post EAGE conference: “Frontier Exploration is back”

The 86th EAGE Annual Conference & Exhibition was held in Aberdeen on 8–11 June 2026, bringing together the full seismic and subsurface supply chain. On 16 June, we hosted **Viridien (ex-CGG) CFO Jerome Serve** for a fireside chat to share feedback from the event and discuss the broader state of the seismic market post-consolidation. Key takeaways include:

- 1. EAGE 2026 was the strongest signal in years.** Record attendance of >6k delegates, with C-suite engagement stepping up vs 2025, underpinned by **reserve replacement and reserve life firmly back on the agenda and reinforced by post-war energy security concerns**. This has not yet flowed into seismic revenues — 2026 budgets were locked pre-war — **with the inflection expected by Viridien in 2027 (Equinor CMD +9% capex a directional positive)**; we see this as a **disciplined multi-year up-cycle, not a repeat of 2010–15**.
- 2. The dominant theme was the return of frontier exploration.** According to Viridien, historically ~80% of activity was near-field, but near-field alone is insufficient to offset falling reserve life; **the last few months show a clear reopening of frontier work, with rising client interest across Uruguay, Equatorial Margin Brazil, Angola, Egypt, Liberia and a broadening GoA**. Majors will need time to rebuild frontier expertise after years of underinvestment according to the company.
- 3. Reprocessing-led demand was the clear second theme and plays directly to seismic companies.** According to Viridien, it's clients' own budgets show more reprocessing in frontier basins, often ahead of drilling, as they look to lock up data early. Behavioural shifts reinforce the trend: prefunding traction is rising, exploration success rates remain low (~20–25%, basin-dependent) so **image quality remains the differentiator, and NOCs increasingly outsource the full image-to-prospect stack to seismic players rather than build it in-house**.
- 4. Vessel availability is not a constraint today according to the company; day rates stable but with upside if capacity tightens.** Streamer fleet consolidated at ~16–17 active vessels (TGS/Shearwater dominant) with stacked capacity reactivatable on a lag; OBN supply is unconstrained but ~4–5x more expensive than streamer-equivalent. Pricing has not increased yet despite consolidation,

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with Viridien expecting streamer cost inflation to pass through into MCL pricing and OBN pricing to fall over time as deployment efficiency improves.

5. AI in seismic is accretive, not substitutive — image quality remains the moat.

Viridien believes that AI cannot solve the physics-based wave equations underpinning imaging; HPC capacity is the structural moat. Its contribution is concentrated in denoising, workflow acceleration and faster delivery, while end-clients increasingly layer their own AI on top of the underlying images, raising rather than commoditising the value of high-quality data. **Disintermediation risk is low, in Viridien's view, as its in-house computing still requires data from the multi-client library, while NOCs and independent players will likely choose to rely on seismic player's specialized capabilities.**

Key takeaways from the fireside chat with Viridien post EAGE conference

EAGE 2026 was the strongest signal in years; exploration is “back, or soon to be back”.

- **Record attendance of >6k delegates** (all-time high per organisers), with EAGE the largest seismic event globally alongside IMAGE in Houston, and all key service players and key clients were present.
- **C-suite engagement stepped up vs 2025, with the improved sentiment underpinned by reserve replacement and reserve life firmly back on the agenda and reinforced by post-war energy-security concerns.**
- This has not yet translated into seismic companies revenues, as 2026 client budgets were locked pre-war, with the **inflection expected by Viridien in 2027 budgets** (Equinor's CMD +9% capex guide a directional positive).

Main themes: return of frontier exploration, with reprocessing-led demand a clear second and regional interest broadening

- **Appetite for genuine frontier work has clearly increased:** According to Viridien, historically ~80% of activity was near-field; near-field exploration continues but is not sufficient to offset the falling reserve life, with it's clients' own budgets showing more reprocessing in frontier basins, often ahead of drilling, as they look to lock up data early. The **last few months show a genuine return to frontier**, per Viridien, but **majors have lost frontier-basin expertise after years of underinvestment and will need time to rebuild it.**
- Viridien flagged **rising client interest across Uruguay (Namibia analogue), Equatorial Margin Brazil (completely unexplored, massive area, Petrobras drilling its first well), Angola (multiple majors back in), Egypt, Liberia, and a broadening US Gulf of America** as more independents return — with MoUs being signed including with countries Viridien had never engaged before.
- Viridien believes its Earth Data library is best positioned in Norway, GoM and Brazil, all currently seeing significant client interest.
- Reprocessing is the near-term workhorse and plays directly into Viridien's edge (proprietary AI + in-house HPC), offering clients a cheaper route to better basin understanding than new acquisition.

Cycle outlook: a disciplined multi-year up-cycle, not a repeat of 2010–15

- Management was explicit and believes that **the industry will not return to the spending exuberance of 10–15 years ago**, when the prior cycle peaked at ~60 active streamer vessels — a level unlikely to be revisited. It believes clients are permanently more shareholder-return focused and selective.
- Management is convinced that **2027 budgets will be set materially higher, marking the inflection point for the new cycle.**
- A sustained, multi-year up-cycle through the decade is plausible, per the company, supported by reserve-replacement pressure, frontier re-engagement, a rising

sovereign MoU pipeline, growing prefunding, and replacement-driven new streamer/OBN technology sales.

Client appetite for seismic is shifting to reprocessing first, with a behavioural shift toward prefunding and outsourcing.

- Viridien said that refunding traction is rising (the ability to assemble prefunded surveys is itself a signal of client intent), and the outsourcing trend is accelerating, with NOCs (e.g. ADNOC) increasingly handing the full image-to-prospect workflow to Viridien rather than building it in-house.
- Exploration success rates remain low (~20–25%, basin-dependent), so image quality remains the differentiator.

Vessel availability is not a constraint today; day rates stable but with potential for upside if capacity tightens

- The streamer fleet is consolidated at ~16–17 active vessels, dominated by TGS and Shearwater (PXGEO, Dubai, Chinese players on the fringe); **no issue sourcing vessels and stacked capacity can be reactivated (with a lag) if the market tightens, according to Viridien.**
- OBN is a different supply curve: no specialised vessels needed and competition small, but absolute price remains ~4–5x more expensive than streamer-equivalent; its clients want more OBN but are price-restrained, and the industry is working to reduce sensor deployment time.
- **Pricing has not increased yet despite consolidation;** if streamer costs rise, Viridien will pass through into MCL pricing, while OBN pricing is expected to fall over time.

AI in seismic is accretive, not substitutive — image quality remains the moat, and Viridien is positioned to capture it.

- At the industry level, Viridien believes that **AI cannot solve the physics-based wave equations that underpin imaging — those remain iterative, HPC-intensive problems — so HPC capacity is still the structural moat.**
- AI's contribution is concentrated in denoising, data cleaning and workflow acceleration, while end-clients increasingly layer their own AI tools on top of seismic images to refine prospectivity — image quality is the binding constraint on the value those tools can extract, which raises (rather than commoditises) the value of high-quality underlying data.
- For Viridien specifically, AI is being deployed across its processing workflow and a new cloud delivery interface now ships terabyte-scale image volumes in ~1 day vs ~4 weeks historically, accelerating client re-engagement.
- Sophisticated clients combine Viridien images with their own interpretation tools; less sophisticated NOCs outsource the full image-to-prospect stack to Viridien, supporting the structural outsourcing trend.
- **Disintermediation risk is low, according to the company, as their in-house computing still requires data from the multi-client library, while NOCs and**

independent players will likely choose to rely on seismic player's specialized capabilities.

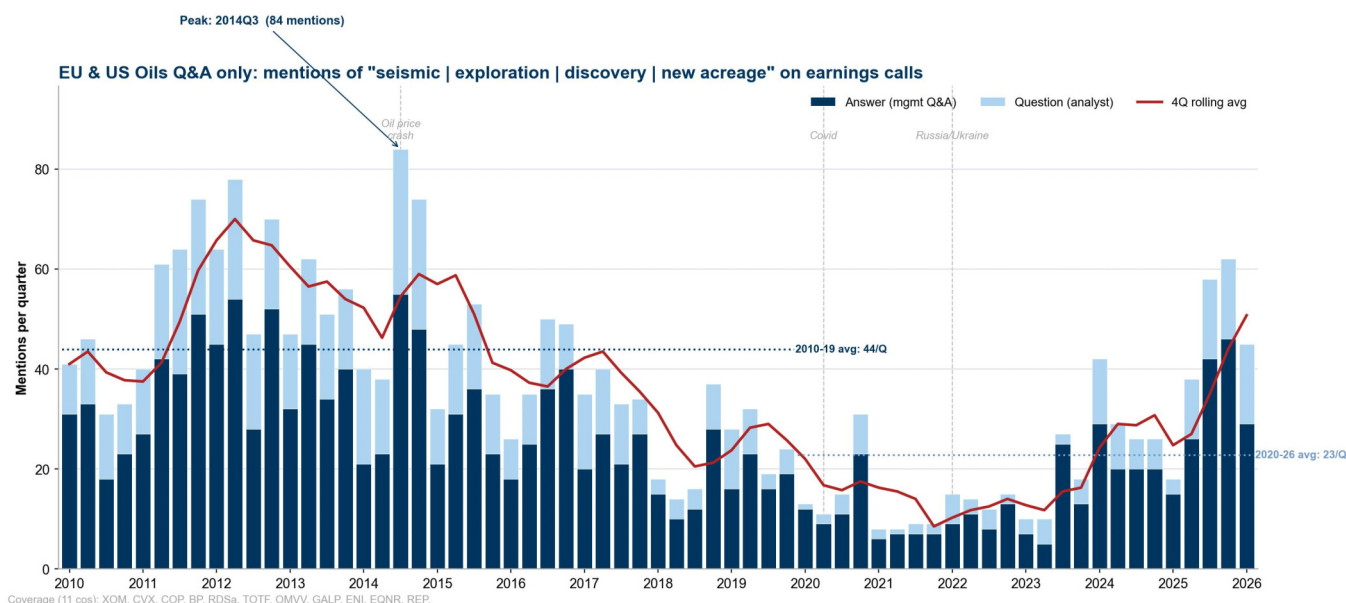
E&P spending cycle: the beginning of a new oil capex upcycle?

The seismic industry is an amplified derivative of the E&P capital expenditure cycle, in our view, with exploration expenses acting as one of the most volatile components of upstream budgets. The sector experienced a severe, multi-year contraction between 2014 and 2021, during which time the global seismic market shrank. The marine seismic industry has seen a **c.85%–90% contraction in streamer fleet capacity** since 2014.

The revival of exploration

We are starting to see early signs of a revival in exploration activity, with both management commentary and analyst questions turning incrementally more constructive on new resource discovery, seismic work and frontier acreage. As shown in [Exhibit 1](#), mentions of exploration-related themes—which troughed in 2020–22 following years of capital discipline—are now clearly inflecting upwards and approaching the high levels of 2014. This shift is consistent with a gradual rebalancing of capital allocation as balance sheets have strengthened and companies look to replenish resource bases after a prolonged period of underinvestment. While still early-cycle, the change in tone suggests that exploration is moving back up the agenda for the majors, supporting a recovery in seismic demand over time.

Exhibit 1: While still early cycle, the change in tone suggests that exploration is moving back up the agenda for the majors, supporting a recovery in seismic demand over time



Source: GS Data Works, Data compiled by Goldman Sachs Global Investment Research

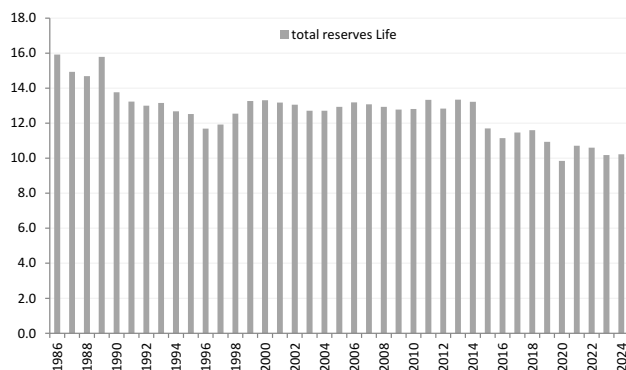
We believe the sector is poised for a significant oil capex upcycle, similar to that of the early 2000s. With OPEC global spare capacity running down, US shale reaching maturity, and reserve life eroded by under-investment, we were already seeing increasing signs of a new oil capex cycle returning, similar to the 2000s. The escalation of geopolitical tensions in the Middle East since late-February has acted as a catalyst to

accelerate and bring forward what we see as a structural capex upcycle coming in 2027E, driven by the following:

1) The industry needs to re-invest in reserve life with more exploration capex: The resource life of Top Projects (recoverable resources/production) has fallen to c.20 years in 2026E from c.55 years in 2012, a 60% decrease over the past decade, as the industry has focused on short-cycle and brownfield (to support lower decline rates in the near term, at the expense of longer reserve life).

Exhibit 2: Industry reserve life was broadly flat at 10.2 years in 2024

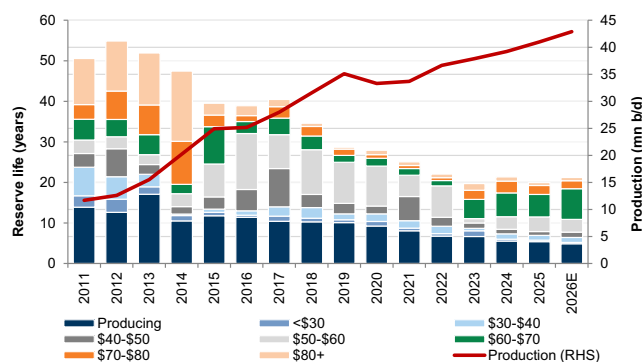
Reserve life (year) per FAS69 data



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 3: Top Projects oil reserve life has fallen 35 years since 2012

Top Projects reserve life, by year of report and breakeven

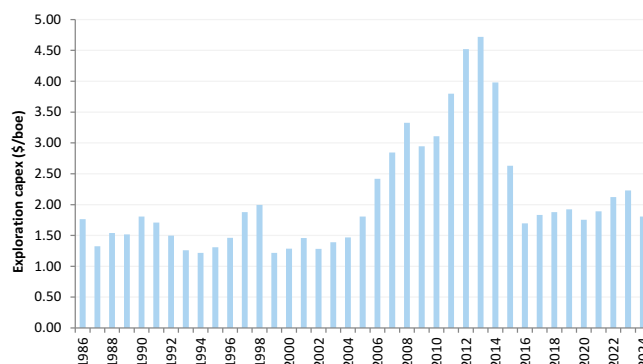


Source: Goldman Sachs Global Investment Research

Conventional discoveries have remained low, with Oil & Gas capex/boe still c.40% below the peak and **exploration down c.60%**, tilting the sources of new reserves increasingly toward revisions and M&A, rather than organic, exploration-led additions. Total capex per barrel continued its downward trend to \$14.7/boe in 2024, down 39% from its \$24.2/boe peak in 2013. In particular, exploration capex was down 59% from its peak at \$1.8/boe in 2014.

Exhibit 4: Exploration capex was down 59% from its peak at \$1.8/boe in 2014

Exploration capex per bboe per FAS69 data



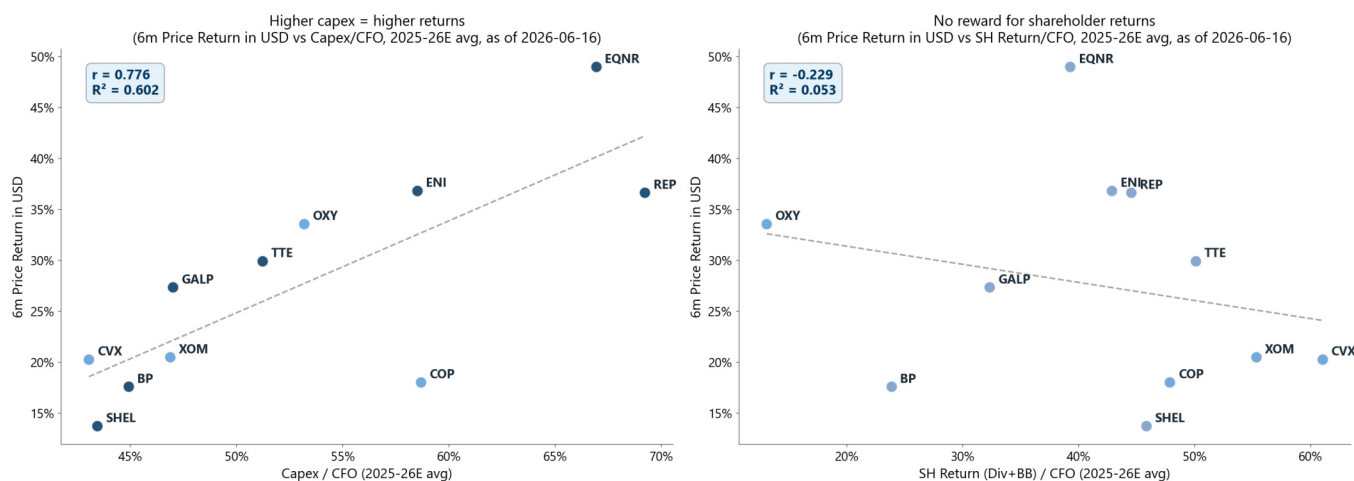
Source: Company data, Goldman Sachs Global Investment Research

2) The market is rewarding reinvestment again: After years of demanding capital discipline and cash returns, the market appears to be shifting back towards rewarding capex and growth. Across our integrated oil universe, companies with higher

reinvestment ratios (Capex/CFO) have outperformed lower-reinvestment peers by **14pp** over the past 6 months. Among the supermajors, TotalEnergies (+30%) has outperformed Shell (+14%) and BP (+18%) in Europe; in the US the relationship is more pronounced over the last 12m while 6m shows a more limited dispersion, with ExxonMobil (+26%/+21%) and Chevron (+23%/+20%). Capex intensity shows a strong positive correlation with 6-month share price performance ($r = +0.78$), while the relationship with shareholder distributions (dividends + buybacks) has been negligible. At the same time, the reinvestment ratio is still **c.15pp** below the long-term average (c.60%/45% of CFO for EU/US Oils vs. historical averages of c.75%/60%).

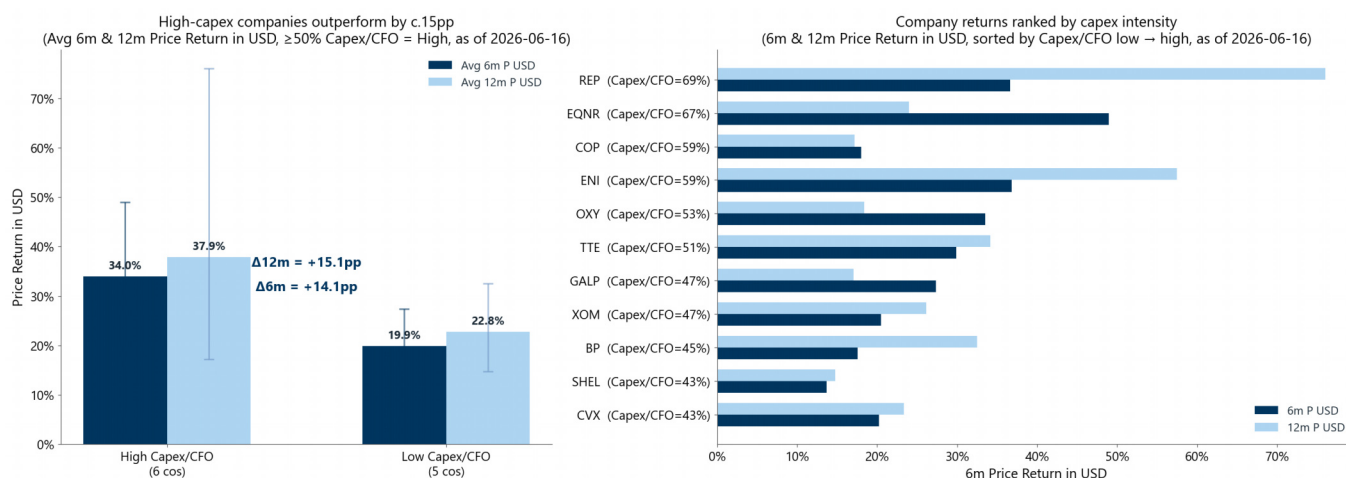
Exhibit 5: Capex reinvestment is being rewarded by the market and has correlated with higher total return in the last 12 months

12-month share price returns in USD



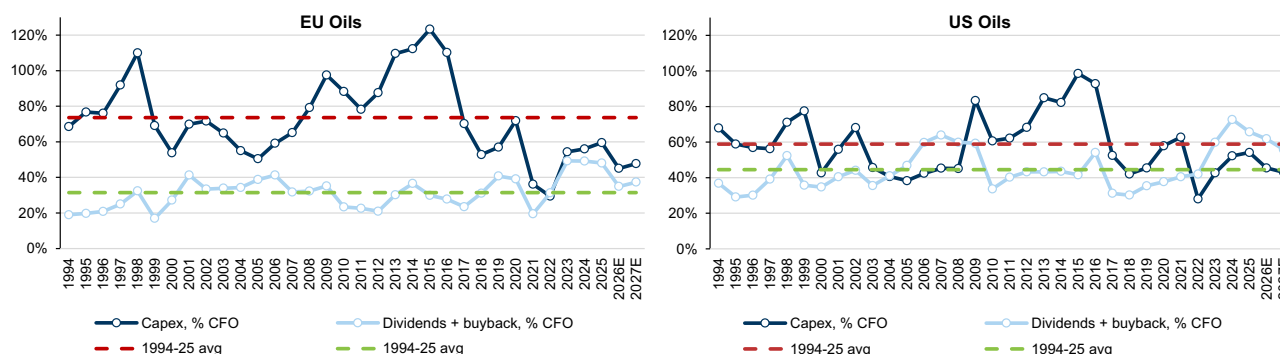
Source: LSEG Data & Analytics, Goldman Sachs Global Investment Research

Exhibit 6: Longer reserve-life companies have outperformed as the market rewarded resource depth



Source: LSEG Data & Analytics, Goldman Sachs Global Investment Research

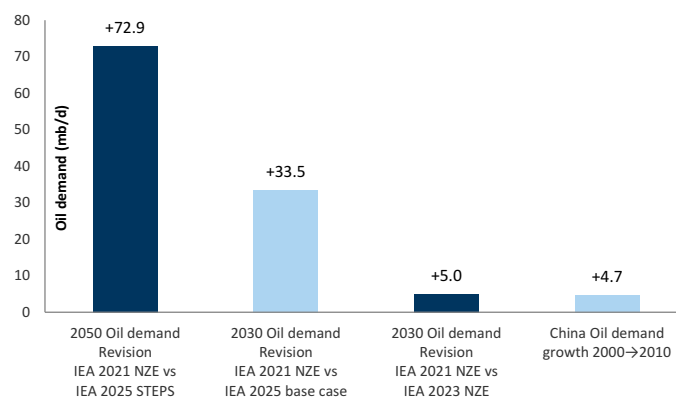
Exhibit 7: Capital allocation remains skewed towards distributions with payout levels >10 pp above history and reinvestment levels remaining c.15 pp below historical norms
 EU Oils vs. US Oils capex intensity and payout, % CFO



Source: Company data, Goldman Sachs Global Investment Research

3) We are seeing structural upward revisions to the long-term oil demand outlook at a time when OPEC spare capacity is at a multi-year low and US shale production is flattening out. Nearly all the world's global crude spare capacity (which our commodities team estimate at 3.7mb/d before the war) is currently constrained by disruptions in the Strait of Hormuz (SoH). However, assuming no large persistent losses in potential oil output, history suggests OPEC is likely to deploy its spare capacity after SoH's reopening to help stabilize tight markets and make up for some production losses during the disruption, when production dropped below the quota. Separately, OPEC+ members will conduct an annual capacity audit in November 2026 to reset 2027 baselines to ensure that the group sets output quotas that are more closely aligned with each country's real capacity. At the same time, the IEA has increased the 2030 oil demand forecast by **5 mn b/d** vs. the **2021** NZE report – equivalent to growth in China demand in the first decade of the 2000s.

Exhibit 8: IEA's NZE +5 mboed upward revision to 2030 oil demand is comparable in scale to China's oil demand growth over 2000-10
 Oil demand (mb/d)



Source: IEA, Statistical Review of World Energy

Competition analysis

Seismic services: Since 2014, the marine seismic industry has undergone a super consolidation phase, moving from eight meaningful global players to three players by

2026, following a prolonged downturn marked by bankruptcies, asset sales and restructuring. Between 2015 and 2022, Dolphin, Polarcus and ION Geophysical withdrew from the market following liquidation or Chapter 11, while CGG and WesternGeco transitioned away from the acquisition models. The consolidation phase accelerated post restructuring: Shearwater acquired the WesternGeco fleet (2018), Polarcus assets and CGG vessels; TGS acquired ION's data library (2022) and Magseis Fairfield (2023), and merged with PGS in July 2024, absorbing its entire fleet; CGG changed its name to Viridien in 2024 and fully exited vessel ownership by early 2025. As a result, ownership of high-end streamer and OBN capacity is now concentrated among TGS and Shearwater, alongside Chinese state-backed COSL/BGP.

This consolidation has been accompanied by a significant contraction in supply. Global active 3D seismic vessel capacity has fallen by c.70%, from c.60 vessels in 2013 to c.17-20 today, with no newbuilds since 2017 and an ageing fleet. The industry has bifurcated into two distinct models: capital-intensive, fully integrated operators (TGS and Shearwater) controlling the vast majority of Western streamer and OBN capacity, and asset-light data and technology players such as Viridien and SLB (WesternGeco) focused on multi-client libraries, imaging and digital solutions. High replacement costs, limited shipyard availability and disciplined capital allocation have materially raised barriers to entry. With exploration activity recovering, this structurally tighter market supports improved utilisation, pricing power and cash generation for the remaining operators, marking a break from the fragmented, excess capacity model that defined the pre-2014 cycle.

Exhibit 9: Seismic services: 2014 vs now

Seismic Services						
2014 Players	Status	Comments	2026 Players	Status	2014 →	Now
CGG	Restructured	renamed Viridien	TGS	Consolidator	8	3
WesternGeco-(SLB)	Merged	with Shearwater	Viridien	Restructured	players	players
TGS-NOPEC	Active		Shearwater	Active	-60% consolidation	
PGS	Merged	with TGS	COSL (seismic)	Active		
ION Geophysical	Restructured	Pivoted				
Dolphin	Bankrupt					
Polarcus	Bankrupt					
COSL (seismic)	Active	CNOOC subsid.				

Source: Company data, Goldman Sachs Global Investment Research

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Reg AC

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